

Lesson 12 GPIO Port Control

This lesson aims to demonstrate on how to utilize the high and low voltage levels of GPIO port on the expansion board.

1. Working Principle

This program controls the high and low voltage levels of a GPIO port on the Raspberry Pi to rotate a fan module. The fan will rotate clockwise for 3s, followed by counterclockwise rotation for another 3s. The source code of the program is located in “/home/pi/board_demo/gpio_control.py”.

```
17 chip = gpiod.Chip("gpiochip4")
18 fanPin1 = chip.get_line(8)
19 fanPin2 = chip.get_line(7)
20
21 fanPin1.request(consumer="pin1", type=gpiod.LINE_REQ_DIR_OUT)
22 fanPin2.request(consumer="pin2", type=gpiod.LINE_REQ_DIR_OUT)
23
24 def set_direction(pin1_value, pin2_value):
25     fanPin1.set_value(pin1_value)
26     fanPin2.set_value(pin2_value)
27
28 def rotate_clockwise():
29     set_direction(1, 0)
30
31 def rotate_counterclockwise():
32     set_direction(0, 1)
33
34 def stop_rotation():
35     set_direction(0, 0)
36
37 def cleanup(signal, frame):
38     stop_rotation()
39     fanPin1.release()
40     fanPin2.release()
41     chip.close()
```

- ① This section provides GPIO pin control through the gpiod module.
- ② Create an instance of the “Chip” class to initialize GPIO.
- ③ The “get_line” method is used to obtain two GPIO pins 8 and 7 from the Raspberry Pi, which are respectively assigned to the variables “fanPin1” and “fanPin2”.
- ④ The “request” method is used to set the direction of both pins to output.

⑤ The “set_direction” function sets the values of “fanPin1” and “fanPin2” to control the rotation direction of the fan.

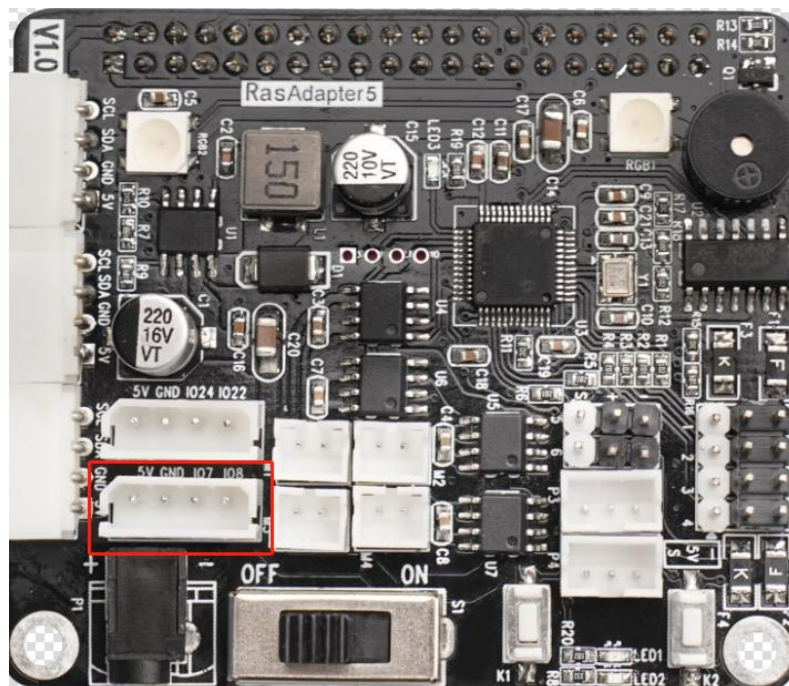
⑥ The “rotate_clockwise” function calls “set_direction” to rotate the fan clockwise.

⑦ The “rotate_counterclockwise” function calls “set_direction” to rotate the fan counterclockwise.

⑧ The “stop_rotation” function calls “set_direction” to stop the rotation of the fan.

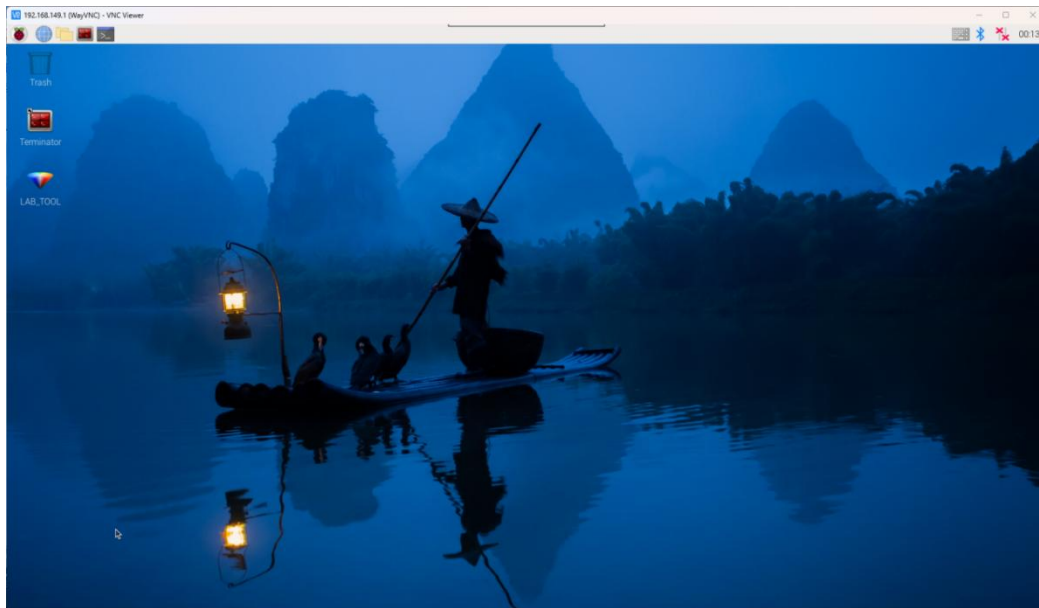
2. Getting Ready

There are 4 GPIO ports on the Raspberry Pi expansion board. Let's use GPIO7 and GPIO8 to control.

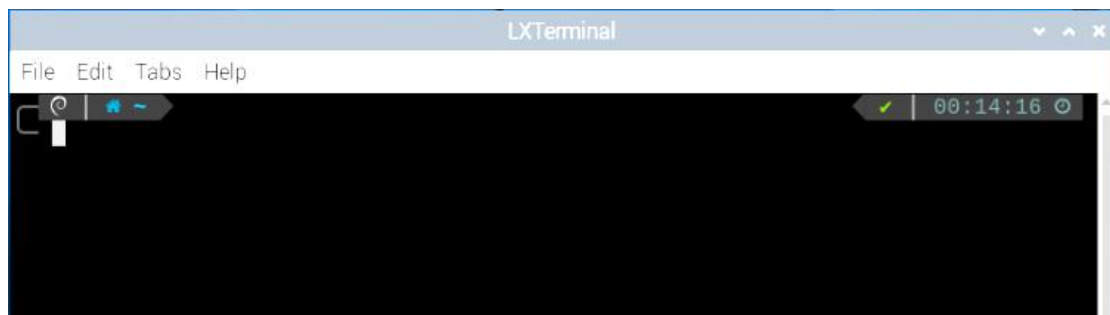


3. Operation Steps

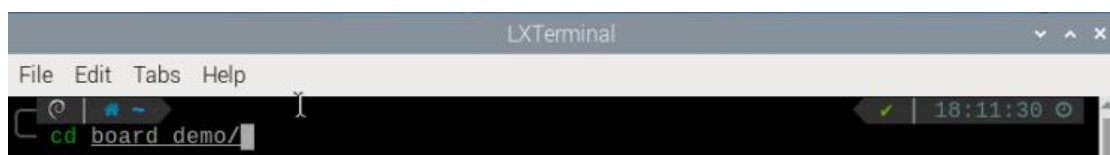
1) Turn the device on and connect the robot through the VNC remote connection tool.



2) Click  in the top left corner of the desktop, or press the shortcut key “Ctrl+Alt+T” to open the command line terminal.



3) Enter the command “cd board_demo/” to navigate to the directory where the demo program is located, then press “Enter”.



Enter the command of running the program “python3 gpio_control.py”, and press “Enter”.

```
pi@raspberrypi:~/board_demo $ python3 gpio_control.py
*****功能:幻尔科技树莓派扩展板, GPIO应用例程(function:Hiwonder Raspberry Pi
expansion board, GPIO application)*****
-----
Official website:https://www.hiwonder.com
Online mall:https://hiwonder.tmall.com
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Tips:
* 按下Ctrl+C可关闭此次程序运行, 若失败请多次尝试! (Press "Ctrl+C" to exit the
program. If it fails, please try again multiple times!)
-----
```

4) Press “Ctrl+c” to close the program. If the program cannot be closed successfully, repeat this operation until it exits.

4. Program Outcome

After the program is executed, the GPIO pins transform between high and low voltage levels to control the fan’s rotation. Adjust the parameters can change the angle place and runtime of different servos to implement the speed control.